INCH-POUND

MS75085E
31 August 2007
SUPERSEDING
MS75085D
16 February 1995

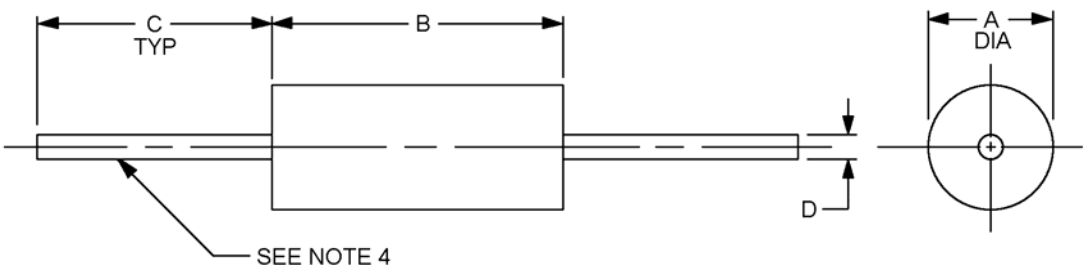
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, (FERRITE CORE), TYPES LT10K078 TO LT10K096, INCLUSIVE

Inactive for new design, after 4 September 1985.
For new design, use MIL-PRF-39010/10.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



LTR	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.085 (2.16)	.105 (2.67)
B	.240 (6.10)	.260 (6.60)
C	1.250 (31.75)	1.625 (41.28)
D	.0185 (0.470)	.0215 (0.546)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their leads.
4. Solderable/weldable lead wire, number 24 AWG.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Weight: 0.0106 ounce, maximum.

Operating temperature range: -55° to +105°C.

Ambient temperature: + 90°C maximum.

Temperature rise: 15°C.

Terminal pull: 5 pounds, minimum.

Altitude: 70,000 feet.

Shock, specified pulse: Method 213 of MIL-STD-202, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: Method 301 of MIL-STD-202, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: Method 105 of MIL-STD-202, test condition C, test voltage 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS75085 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

Dash Number MS75085	Type Designa-tion	Inductance (μH) ±10%	Q Min.	Test Frequency (L and Q) (MHz)	SRF Minimum (MHz)	DC resistance (25°C) max (Ohms)	Rated DC current, (mA)
-1	LT10K078	33	45	2.5	24	3.4	130
-2	LT10K079	39	45	2.5	22	3.6	125
-3	LT10K080	47	45	2.5	20	4.5	110
-4	LT10K081	56	45	2.5	18	5.7	100
-5	LT10K082	68	50	2.5	15	6.7	92
-6	LT10K083	82	50	2.5	14	7.3	88
-7	LT10K084	100	50	2.5	13	8	84
-8	LT10K085	120	30	.79	12	13	66
-9	LT10K086	150	30	.79	11	15	61
-10	LT10K087	180	30	.79	10	17	57
-11	LT10K088	220	30	.79	9	21	52
-12	LT10K089	270	30	.79	8	25	47
-13	LT10K090	330	30	.79	7	28	45
-14	LT10K091	390	30	.79	6.5	35	40
-15	LT10K092	470	30	.79	6	42	36
-16	LT10K093	560	30	.79	5	46	35
-17	LT10K094	680	30	.79	4	60	30
-18	LT10K095	820	30	.79	3.8	65	29
-19	LT10K096	1000	30	.79	3.4	72	28

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±10	±(5% +.001 ohm)	15	-20
Group IV	±5	±(2% +.001 ohm)	-10	-15
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II	±5	±(2% +.001 ohm)	-10	-15
Subgroup III	±10	±(5% +.001 ohm)	-15	-20

1/ Test fixture allowance of +.01μH shall be added to all change in inductance limits ±(_ percent +.01 μH).

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Application notes:

1. After the overload test is performed, a period of 24 hours shall elapse prior to taking electrical characteristics (final) measurements.
2. The polarization voltage during moisture the moisture resistance tests is applied with the positive lead connected to the coil terminals tied together, and the negative lead connected to the metal strap.
3. DC resistance shall be the last measurement taken in the electrical characteristics test sequence.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-15305
MIL-STD-202

Custodians:
Army – CR
Navy - EC
Air Force - 11
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2007-038

Review activities:
Army – AR, CR4, MI
Navy – AS, CG, MC, OS, SH
Air Force – 19, 99

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